

LLM-driven heuristic evolution

Feb 03 2026

CS395T

Administrivia

- Schedule change:
 - Feb 17: Heimdall paper (Learned I/O admission)
 - Feb 19: WiscKey to Bourbon (Learned Index)
- Project schedule:
 - Feb 26: Project ideas
 - Mar 12: Proposal due
 - Apr 21 and 23: Presentations
 - Apr 27: Final reports due

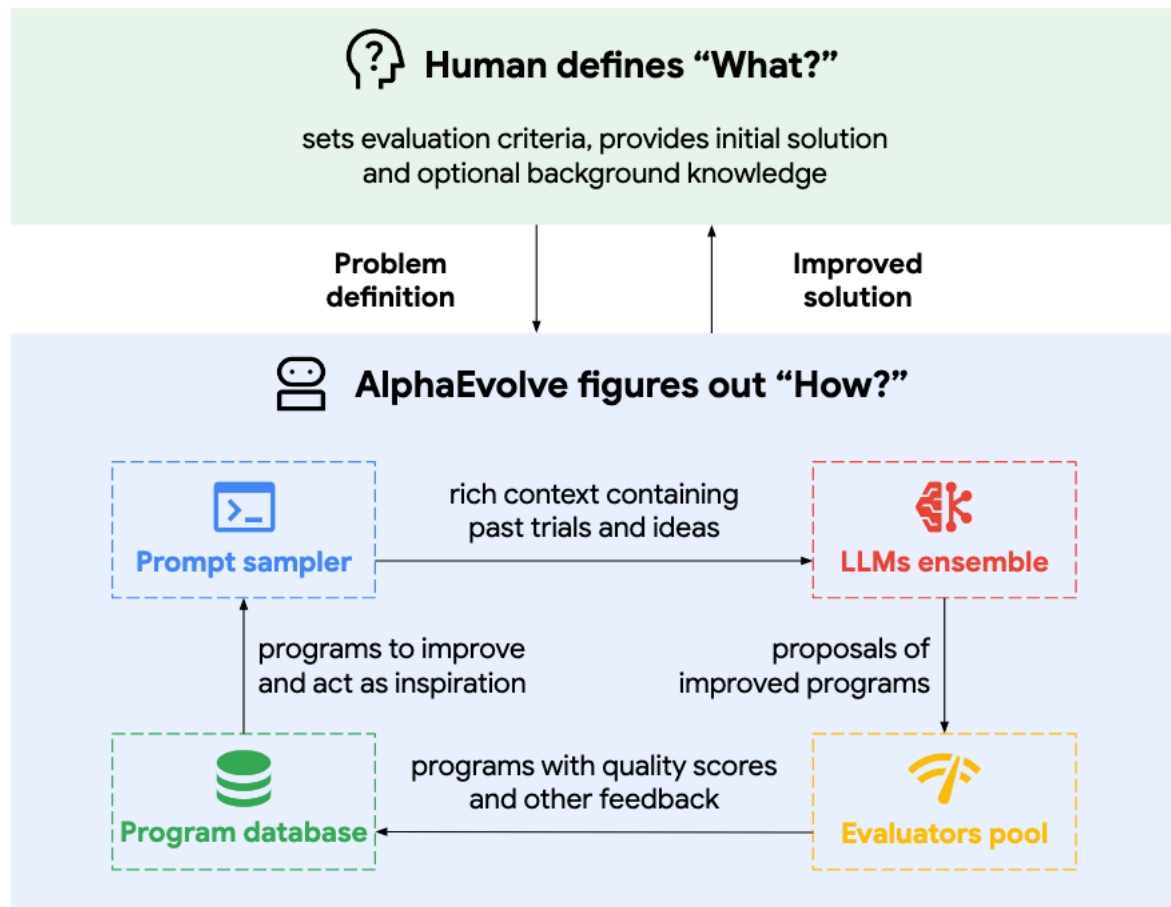
Google Deepmind: FunSearch

- Can LLMs discover new knowledge?
 - Try to solve open problems in math or science
- Discovery: Solution to cap set problem
 - Better algorithm for bin-packing
- Interpretable: Can understand how solutions are constructed

FunSearch

- User provides
 - Problem description
 - Seed program(s)
 - Evaluator framework
- LLM generates new programs, evaluates, adds to program database

AlphaEvolve



AlphaEvolve

<i>FunSearch</i> [83]	<i>AlphaEvolve</i>
evolves single function	evolves entire code file
evolves up to 10-20 lines of code	evolves up to hundreds of lines of code
evolves code in Python	evolves any language
needs fast evaluation (≤ 20 min on 1 CPU)	can evaluate for hours, in parallel, on accelerators
millions of LLM samples used	thousands of LLM samples suffice
small LLMs used; no benefit from larger	benefits from SOTA LLMs
minimal context (only previous solutions)	rich context and feedback in prompts
optimizes single metric	can simultaneously optimize multiple metrics

Open-source variants: OpenEvolve and ShinkaEvolve

Key controllables

- Feedback format and metrics
- Exploration vs exploitation
- Minor edits vs full rewrites

Evolving system heuristics

[1] "Let the Barbarians In: How AI Can Accelerate Systems Performance Research"

[2] "Glia: A Human-Inspired AI for Automated Systems Design and Optimization"

Discussion

1. Why/how is Vulcan able to produce better heuristics? Have we achieved AGI?
2. What are the challenges of deploying Vulcan-evolved heuristics in real-world systems?

Title: [Lec 6] Discussion